

On the Static and Dynamic Motion of Liquid Motion on Solid Surfaces – a RISE Worldwide Internship

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1 Introduction

The onset of liquid droplets on solid surfaces is a phenomenon that is witnessed in every part of our lives and numerous industrial processes. In particular, the application of microfluidic technology has played an ever larger role in the digital high-precision industry. Conversely, the missing consensus on measurement techniques for surface properties and discrepancies between literature values suggest that the solid-liquid interface is not fully understood yet. [1]

Therefore, this work aims at perform precise, repeatable, and unambiguous measurements of contact parameters in dependence on the surface geometry, wettability, and roughness. Specifically, we measure the contact line of water droplets on cylindrical surfaces and quantify its dependence on the surface curvature K , the drop volume V , and the choice of the substrate.

(what has been achieved)

2 About the Internship

The frame of this research project is a six-week internship in the Institute of Applied Mechan-

ics at the National Taiwan University (NTU) in Taipeh, Taiwan. During the project I was integrated into the research of group of professor An-Bang Wang and advised by PhD student Bharanitharan Jambulingan. The week was structured by three group meetings on Monday, Wednesday, and Friday, where we exchanged thoughts on our experiments and discussed new results. I worked in the laboratory from 9:00 to 18:00 except for lunch time, where we would go out with the research group. I really appreciated the matey and cooperative atmosphere as well as all the free time that we could spend together. I learnt a lot about pratical laboratory work, improvisation of solutions, but most of all about the taiwanese culture and the friends I made along the way. Thank you.

3 Theory

When a liquid droplet is placed on a solid surface, its exact geometry generally depends on the choice of liquid and substrate, the droplet volume, and the substrate geometry. The one-dimensional line, where solid, liquid, and surrounding gas meet, is called the **triple-phase boundary** or **contact line**. The inner angle between substrate and droplet surface tangent at the contact line is called the **contact angle**

θ (Fig. 1). For a given droplet volume V , the geometry of the droplet is determined by its contact line and contact angles.

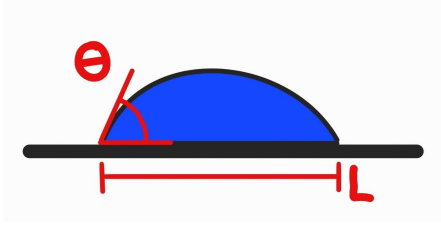


Figure 1: Sketch of applied droplet with contact angle θ and contact length L .

For an ideal combination of liquid and substrate as portrayed by Young’s equation, the **equilibrium contact angle** θ_Y is a characteristic quantity, such that any deformation of the droplet eventually returns to the equilibrium state, where $\theta = \theta_Y$.

However, due to the microscopic and molecular structure of solid surfaces, liquid drops experience pinning effects that induce deviations to the equilibrium contact angle θ_Y , which depend on the history of the droplet: A carefully placed droplet evolves significantly larger contact angles than a droplet that was pressed on the surface. The phenomenon is known as **contact angle hysteresis** $\Delta\theta$ and quantified by the difference in advancing and receding contact angles $\theta_A - \theta_R$ (Fig. 2).

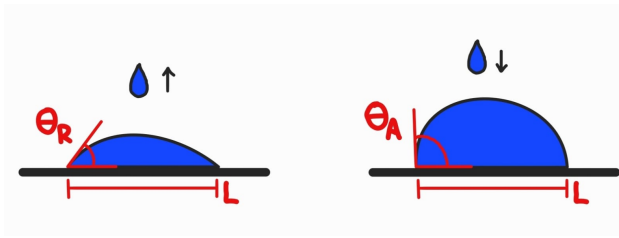


Figure 2: Sketch for measurement of θ_R (left) and θ_A (right) for a given droplet volume V .

Place a droplet on a substrate and enlarge the droplet by slowly supplying it with liquid. Due to static forces, the contact line resists the expansion and the contact angle grows. At

the critical point, before the contact line finally expands, the contact angle reaches the **advancing contact angle** θ_A (Fig. 2, right). Similarly, the **receding contact angle** θ_R is reached at the point where retracting more liquid would make the contact line shrink (Fig. 2, left). The advancing and receding contact angles measure the surface’s repellency, its “resistance to wetting out”.

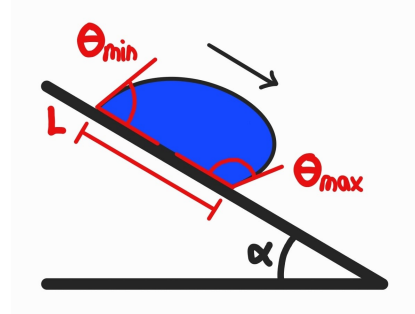


Figure 3: Sketch of the sliding-angle experiment with sliding angle α and contact angles $\theta_{\max}$ & $\theta_{\min}$.

Another material property closely related to repellency is drop mobility, the resistance of a surface to drop movement. It is measured by placing a droplet on a flat surface and tilting it until the critical angle, where the drop starts moving (Fig. 3). The critical inclination angle, at which macroscopic movement of the droplet emerges, is called the **sliding angle** α . In this maximal inclined state, where the droplet is about to slide, the front contact angle is maximized while the rear contact angle is minimized. Thus, they are called the maximum $\theta_{\max}$ and minimum contact angles $\theta_{\min}$. It is noteworthy that despite the close resemblance of $\theta_{\max} \sim \theta_A$ and $\theta_{\min} \sim \theta_R$ for some liquid-substrate combinations, they are generally different and do not measure the same material property (repellency $\neq$ drop mobility).

Ultimately, to make quantitative predictions about the dynamic behaviour of a droplet on a solid surface, we need to understand the ad-

hesion force F that acts on the solid-liquid interface. The **static adhesion force** F_S is the force needed to initiate the movement of a resting droplet over a surface [2]:

$$F_S = \frac{24}{\pi^3} \gamma d (\cos \theta_{\min} - \cos \theta_{\max}) \quad (1)$$

Here, γ denotes the surface tension and d can be approximated by the average diameter of the contact line.

Once the droplet is moving at a constant velocity v a smaller **dynamic adhesion force** $F_D(v)$ is needed to keep the droplet moving [2]:

$$F_D = \frac{24}{\pi^3} \gamma d (\cos \theta_R(v) - \cos \theta_A(v)) \quad (2)$$

Here, $\theta_R(v)$ and $\theta_A(v)$ denote the rear and front contact angles of the droplet moving at a constant velocity v .

As initially stated, this work will investigate the influence of substrate geometry. The simplest adaption we can make here is therefore the use of a cylindrical substrate with diameter D instead of a flat surface ($D \rightarrow \infty$). A droplet placed on cylinder has two axes of symmetry with different contact lengths. In the following, we define these contact lengths via the cylinder axis as L and W as portrayed in Fig. 3.

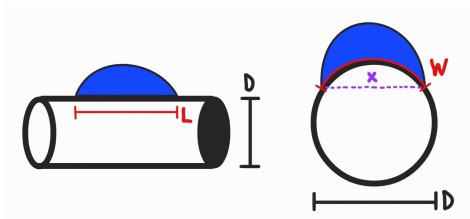


Figure 4: Definition of the perpendicular contact lengths L & W on a cylinder with diameter D . In practice the distance x is measured to calculate W .

Notice that, while L describes a straight line, W wraps around the cylinder and cannot be accurately measured straight away. For that

reason, we measure the straight distance x instead (see Fig. 4), such that

$$W = D \cdot \arcsin\left(\frac{x}{D}\right) \quad (3)$$

Furhtermore, we need a ruler to compare droplet volumes V to measured contact lengths L and contact widths W . As such we define the **reference contact length** $L_0(V)$ as the diameter of a half sphere with volume V :

$$L_0(V) = \sqrt[3]{12 \cdot V / \pi} \quad (4)$$

Droplets with high contact length $L/L_0 < 1$ for their volume V are “flattened”, while droplets with smaller contact lengths $L/L_0 > 1$ are “rounded”. We hypothesize the proportionality

$$L = k_L \cdot L_0(V) \quad (5)$$

$$W = k_W \cdot L_0(V) \quad (6)$$

where k is constant with respect to the droplet volume V . By measuring k on cylinders of different diameters D , we hope to find line shapes $k_L(D)$ and $k_W(D)$ that uniquely characterize substrate-liquid combinations.

Lastly, we consider the **capillary length** L_c as the characteristic length where the two competing forces on a droplet equalize: Below the capillary length ($L/L_c < 1$) surface tension dominates and pulls the droplet together to minimize surface energy. Above the capillary length ($L/L_c > 1$), gravity dominates and deforms droplets to minimize potential energy. With $L_c \approx 2.71$ mm for the water-air interface [3], the experiments investigate this critical region in between.

4 Experimental Setup

To investigate the influence of the substrate geometry on static and dynamic observables, we place water droplets on PMMA cylinders with diameters $D \in \{2.19(2), 3.99(2), 5.15(4), 10.2(2)\}$ mm, using a high-precision mechanical pipette by *Sartorius* for $V \in [2.0, 20.0]$ μ l.

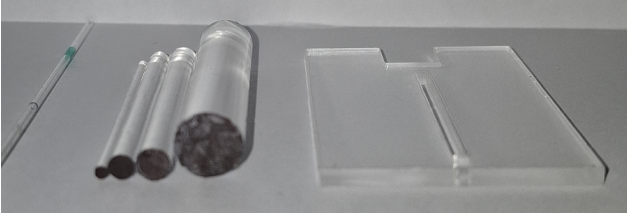


Figure 5: Photo of PMMA samples and fixation plate. The cylinders' back sides are painted black to enhance contrast for measurements of W .

The cylinder is fixed on a movable platform using rubber bands and the observation is supported by a white backlight. All experiments use twice-boiled tap water for the liquid droplets.

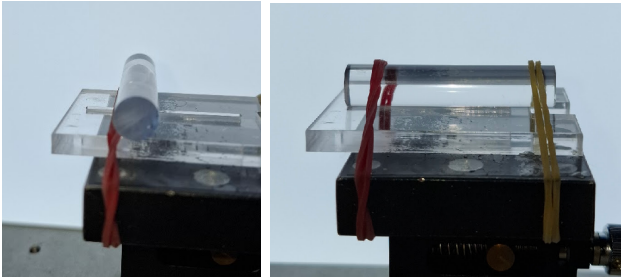


Figure 6: Photo from the camera perspective to measure $W(x)$ (left) and L (right) on the 10 mm cylinder fixed by rubber bands.

Finally, the photography is performed by the *MEMRECAM GX-1* camera (C Mount Depth 9 mm) with a narrow lens and fixed focus for upto 4x magnification. The high-speed camera is operated by the software *MLink* and set to 50 FPS and 250 Shutter. The image format is set to 484x480. To analyze the snapshot, the distances L and x are measured in pixels using the software *ImageJ*. After each measurement, a pipe of known width is placed in the focus of the length to gauge the pixel size for a given magnification.



Figure 7: Photo of the high-speed camera (top) and the 4x magnification lens (bottom).

Between measurements, the sample cylinders are dried and cleaned from dust using *KimwipesTM* for professional use. While the humidity in the laboratory cannot not be controlled accurately, the room temperature during experimental conduction lies between 25 °C and 29 °C. The backlight is not expected to contribute meaningfully to evaporation as it is not focused.

5 Measurement

For a measurement on a given cylinder D for an array of droplet volumes V , the cylinder is cleaned using the wipes and fixed on the platform using rubber bands. Measurements of W in the front position (Fig. 6, left) are performed separately from measurements of L in the side position (Fig. 6, right) on different droplets. Therefore, no pair of L and W belong to the same droplet. To make up for this lack in correspondance, each measurement is performed 4 to 5 times and averaged over.

To place the droplets on the cylinder, one of two procedures is used:

1. The full volume of the droplet is placed on the cylinder as once, measured,

cleaned off, and replaced by the next droplet volume of interest. This method is precise, but slow as the focus has to be readjusted each time. It is used for small cylinders $D < 4$ mm.

2. A 2 mm droplet is placed, measured, and incremented to fit the next droplet volume of interest. The incrementation and measurement is repeated until the droplet becomes unstable and slides off the cylinder. This method is fast, but accumulates error due to the repeated manipulation of the droplet. It is used for large cylinders $D \geq 4$ mm.

It is assumed that the cylinder substrates do not “memorize” the exposure to previously applied droplets such that no additional waiting time is needed between measurements.

Figure 8 shows an exemplary measurement on the 2 mm cylinder, using the procedure (1). On PMMA, the experiment was repeated for droplet sizes 2.2 mm, 2.4 mm, 2.6 mm, 2.9 mm, and 3.2 mm. Due to the limitations of the pipette, drop volumes below 2 mm were inaccessible.

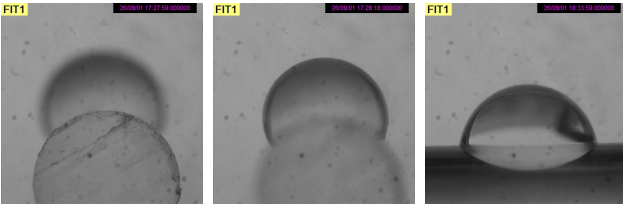


Figure 8: Measurement of a 2 μ L droplet on the 2 mm cylinder from the front (left, middle) and side perspective (right).

On the 4 mm cylinder (Fig. 9), procedure (2) was used, incrementing a 2 mm droplet to 4 mm, 6 mm, 8 mm, 10 mm, and 12 mm. The 12 mm measurement, however, showed repeated instability and therefore cannot be analyzed.

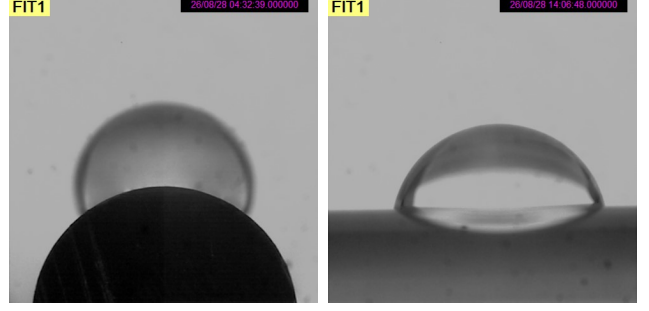


Figure 9: Measurement of a 6 μ L droplet on the 4 mm cylinder from the front (left) and side perspective (right).

Similarly, on the 10 mm cylinder (Fig. 10), procedure (2) was used, incrementing a 2 mm droplet to 4 mm, 8 mm, 12 mm, 16 mm, 20 mm, and 30 mm. The 30 mm, however, showed repeated instability and therefore cannot be analyzed.

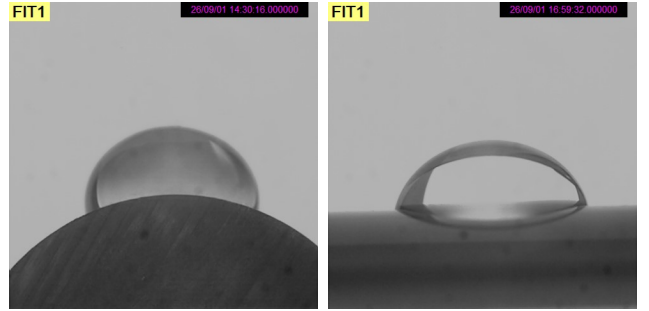


Figure 10: Measurement of a 16 μ L droplet on the 10 mm cylinder from the front (left) and side perspective (right).

Note that due to the short focus length of the camera, it was not always possible to focus the droplet’s silhouette and the cylinder at the same time. In these cases, the focus on the cylinder is preferred as it showed to be more precise and reliable than focussing the droplets individually.

We assume a relative uncertainty of 5% for all lengths measured in the images (L , x , gauge pipe) due to the limitations of the camera focus and image quality. Moreover, this uncertainty is meant to compensate for small deviations in droplet volume and induced vibrations

during droplet placement. Generally speaking, there are countless possible sources of systematic error such as changes in room temperature and humidity, pollution of the sample and liquid, inaccurate camera placement, and evaporation. We nevertheless believe that the dominant source of uncertainty is caused by the droplet placement and sample impurity.

6 Results

7 Conclusion

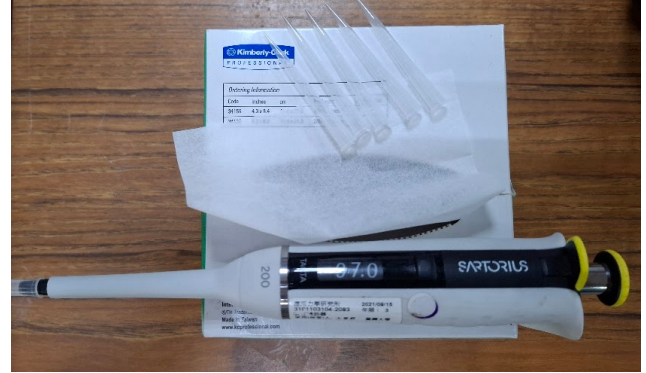


Figure 11: Photo of the 2 μL to 20 μL pipette with *KG1213-L* Low Binding Pipette Tips by Kirgen.

References

- [1] “Project Description for RISE Internship”. 2026.
- [2] D. W. Pilat et al. “Dynamic Measurement of the Force Required to Move a Liquid Drop on a Solid Surface”. In: *Langmuir* 28.49 (Nov. 2012), pp. 16812–16820. ISSN: 0743-7463. DOI: [10.1021/la3041067](https://doi.org/10.1021/la3041067). URL: <https://doi.org/10.1021/la3041067>.
- [3] Bastian E. Rapp. “Capillarity”. In: *Microfluidics: Modeling, Mechanics and Mathematics*. Elsevier, 2017. Chap. 21, pp. 421–442. ISBN: 978-1-4557-3141-1. DOI: [10.1016/B978-1-4557-3141-1.50021-6](https://doi.org/10.1016/B978-1-4557-3141-1.50021-6).

A Supplementary Material

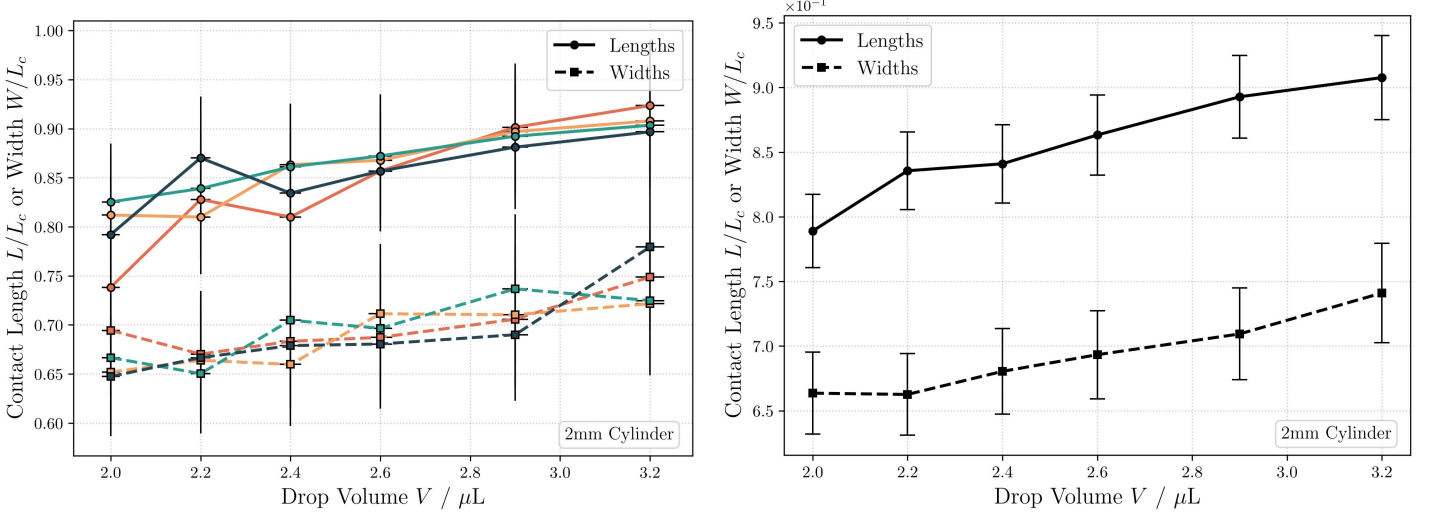


Figure 12: Measurements of contact length L and width W on the 2 mm PMMA cylinder in raw (left) and averaged (right). The contact length L along the cylinder axis always exceeds the width W , however, the region of interest for $V < 2 \mu\text{L}$ could not be measured in practice.

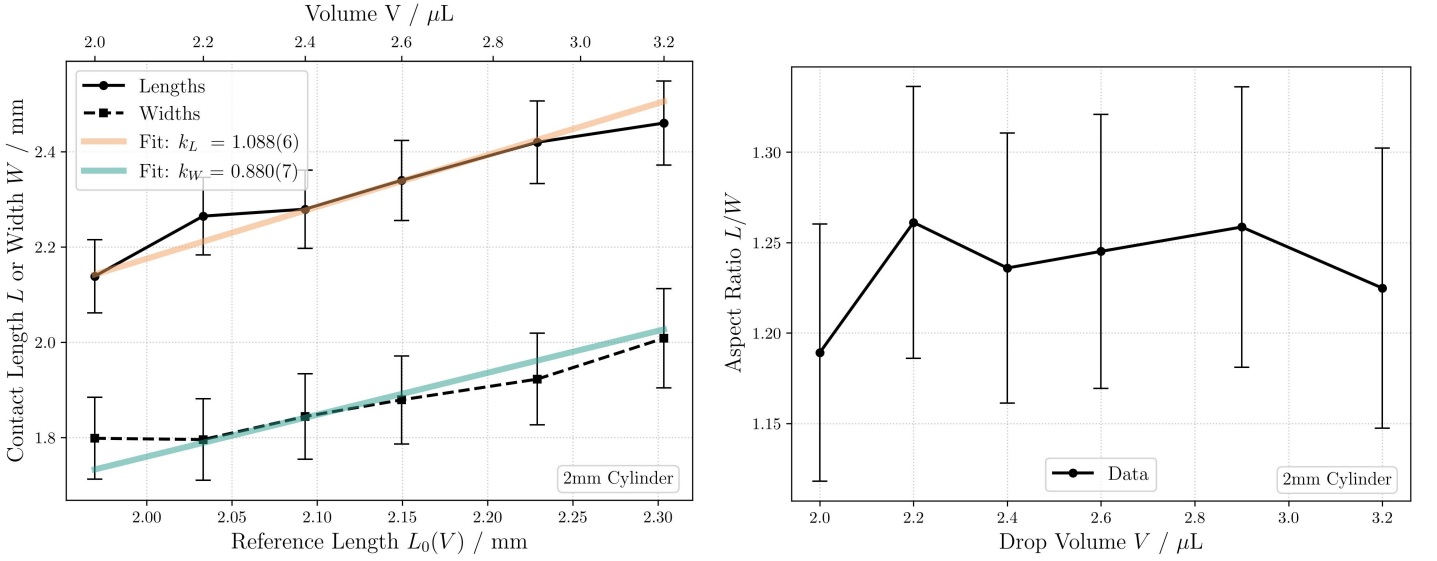


Figure 13: Contact length and width against the reference length $L_0(V)$ with linear fits according to Eq. (5) on the left. Even though the half-sphere model fits the data, $\chi^2 \ll 1$ suggest low significance. Their aspect ratios L/W (right) on the 2 mm PMMA cylinder does not deliver a clear result due to large uncertainties.

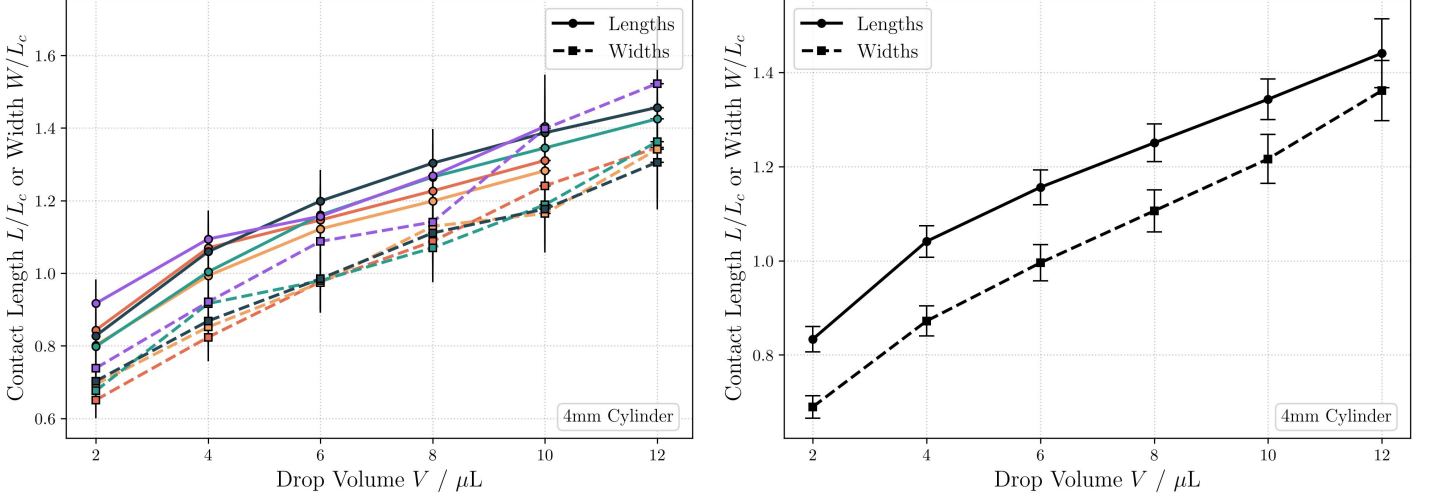


Figure 14: Measurements of contact length L and width W on the 4mm PMMA cylinder in raw (left) and averaged (right). Due to observed instability, the 12 μL data points are not used in the analysis. L and W seem to converge, as gravity starts dominating over the surface tension for $L/L_c > 1$.

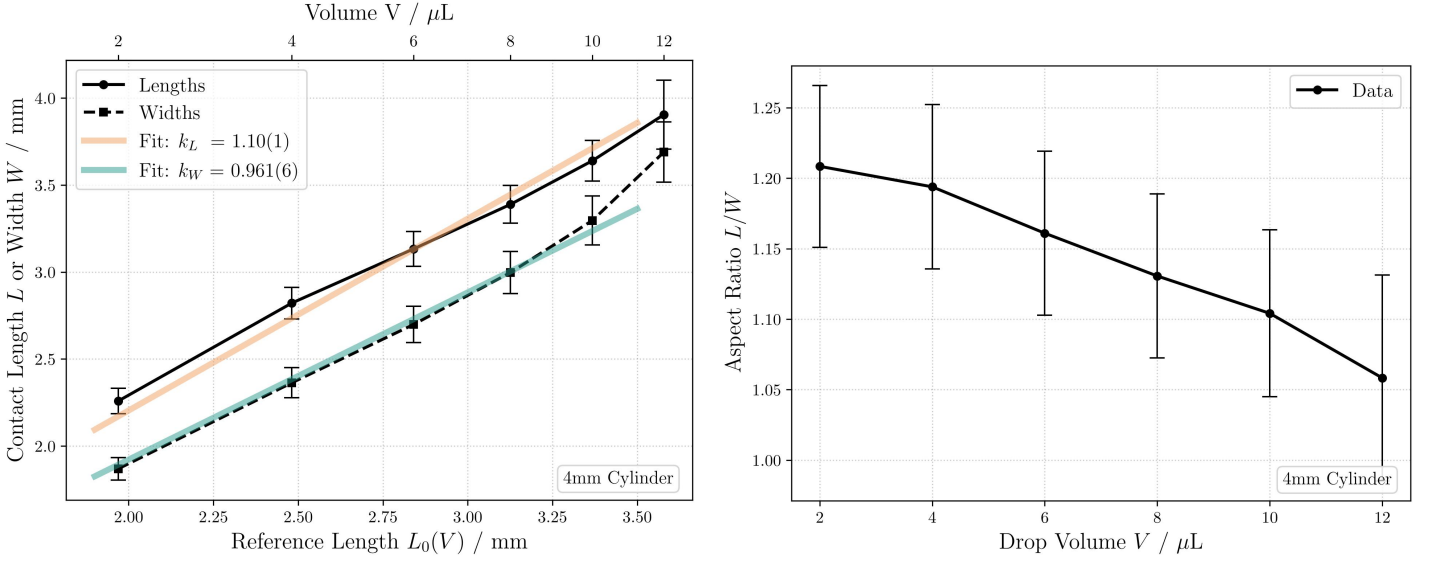


Figure 15: Contact length and width against the reference length $L_0(V)$ with linear fits according to Eq. (5) on the left. Again, the linear model is confirmed with high flexibility ($\chi^2 \ll 1$) and $L > W$ for all V . Their aspect ratios L/W (right) on the 4mm PMMA cylinder trends visibly down to 1, where the droplet width will exceed its length.

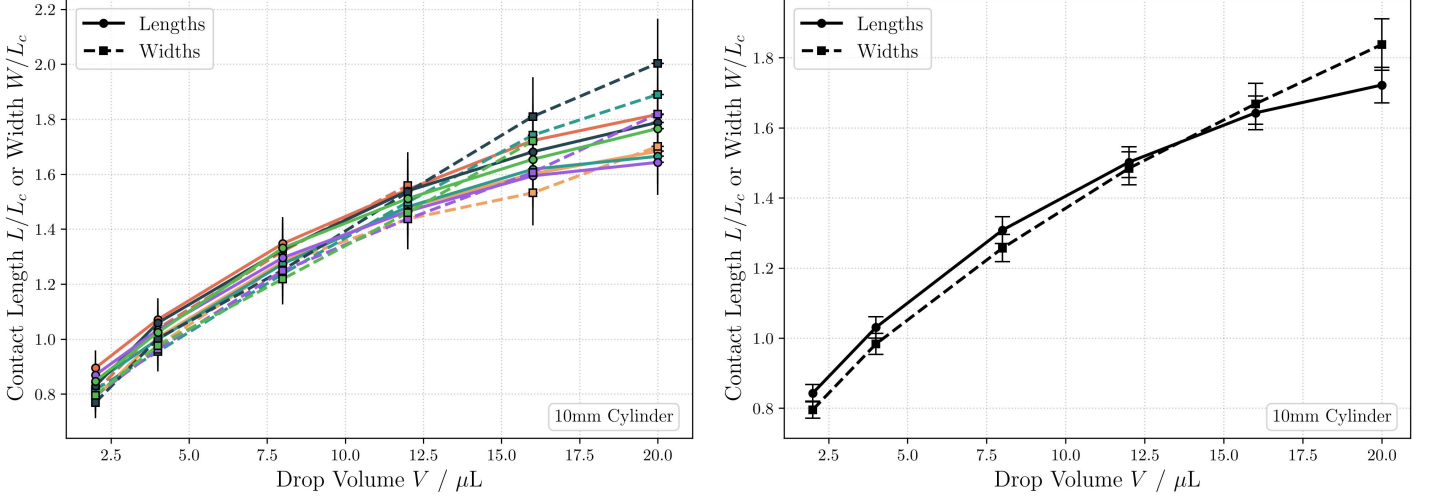


Figure 16: Measurements of contact length L and width W on the 10 mm PMMA cylinder in raw (left) and averaged (right). The length L is overtaking by the width W at $13.5 \mu\text{L}$, indicating the domination of gravity over surface tension ($L/L_c > 1.5$). Due to observed instability, the $20 \mu\text{L}$ data points are not used in the quantitative analysis.

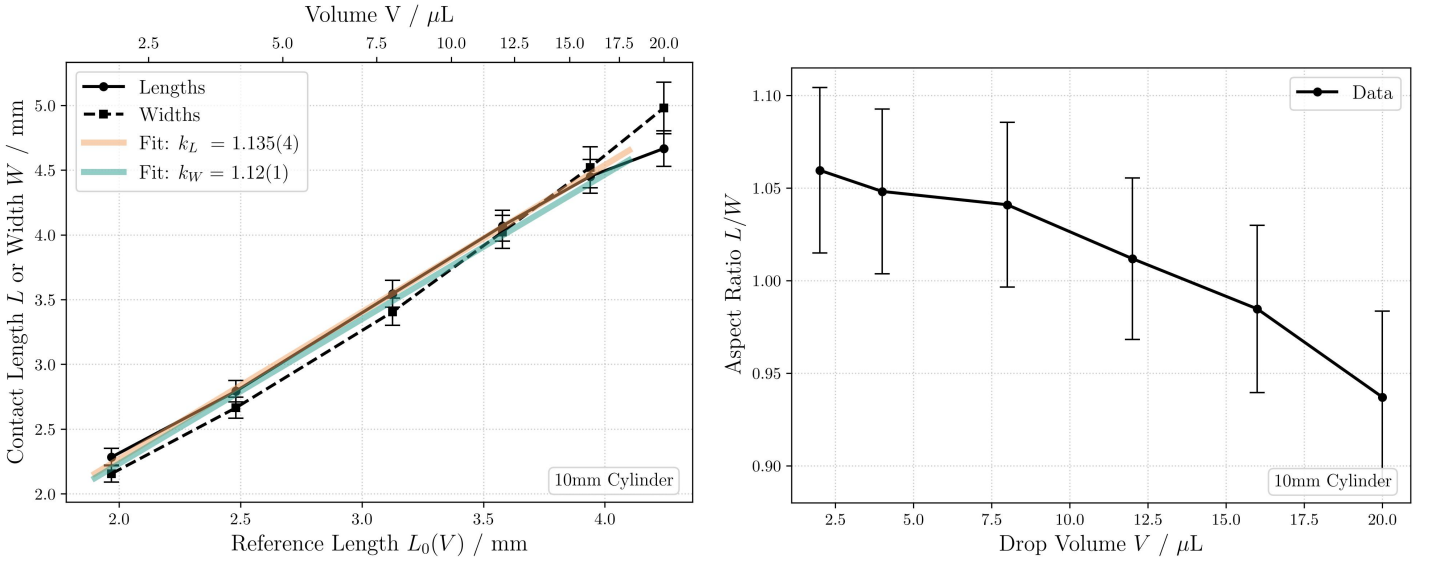


Figure 17: Contact length and width against the reference length $L_0(V)$ with linear fits according to Eq. (5) on the left. As before, the linear model is confirmed with high flexibility ($\chi^2 \ll 1$) and $k_L, k_W > 1$ indicates the “rounded” drop geometry. Their aspect ratios L/W (right) on the 10 mm PMMA cylinder confirm the overtaking of gravity through the decrease in L/W down to below 0.95.